

Computer Graphics (CS7.302)

Line and Polygon Clipping

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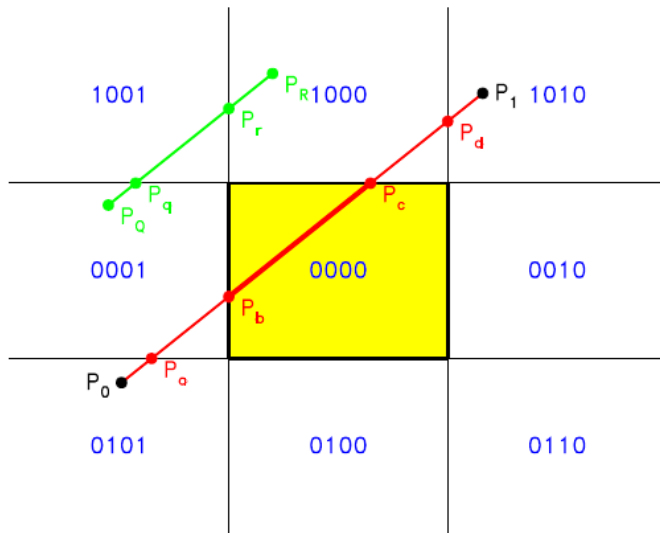
Apr 22nd, 2026

Clipping

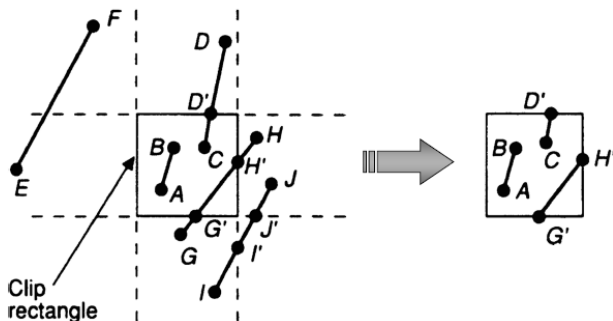
Algorithms

- Brute force
- Cohen Sutherland
- Liang Barsky

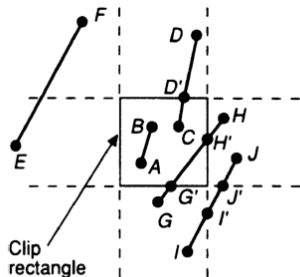
Cohen Sutherland



Cohen Sutherland



Cohen Sutherland



Line	Code 1	Code 2	$C1 \wedge C2$	$C1 \vee C2$
$A \Rightarrow B$	0000	0000	0000	0000
$C \Rightarrow D$	0000	1000	0000	1000
$E \Rightarrow F$	0001	1001	0001	1001
$G \Rightarrow H$	0100	0010	0000	0110
$I \Rightarrow J$	0100	0010	0000	0110

	x_{min}	x_{max}	
1001	1000	1010	
0001	0000	0010	y_{max}
0101	0100	0110	y_{min}

- Trivial Accept $C_1 | C_2 = 0$
- Trivial Reject $C_1 \& C_2 \neq 0$

- Define clip code of a point as four bit value TBRL, where

$$T = \begin{cases} 1, y > y_{max} \\ 0, y \leq y_{max} \end{cases} \quad B = \begin{cases} 1, y < y_{min} \\ 0, y \geq y_{min} \end{cases}$$

$$R = \begin{cases} 1, x > x_{max} \\ 0, x \leq x_{max} \end{cases} \quad L = \begin{cases} 1, x < x_{min} \\ 0, x \geq x_{min} \end{cases}$$

- Let C_0 and C_1 be the TBRL clip code for P_0 and P_1 , respectively.

Then

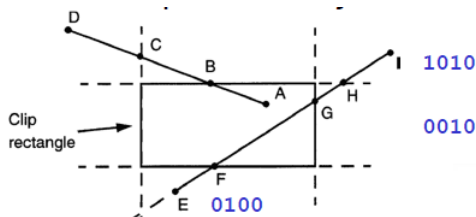
- 1 If $C_0 = 0$ and $C_1 = 0$ the line is completely inside the domain;
- 2 If C_0 and C_1 have one or more bits in common, the line is completely outside the domain;
- 3 If any of the bits in C_0 or C_1 are set, clip to the corresponding edge as follows, updating the end points and clip codes as appropriate

$$T : x = x_0 + (y_{max} - y_0) \cdot \frac{x_1 - x_0}{y_1 - y_0} \quad B : x = x_0 + (y_{min} - y_0) \cdot \frac{x_1 - x_0}{y_1 - y_0}$$

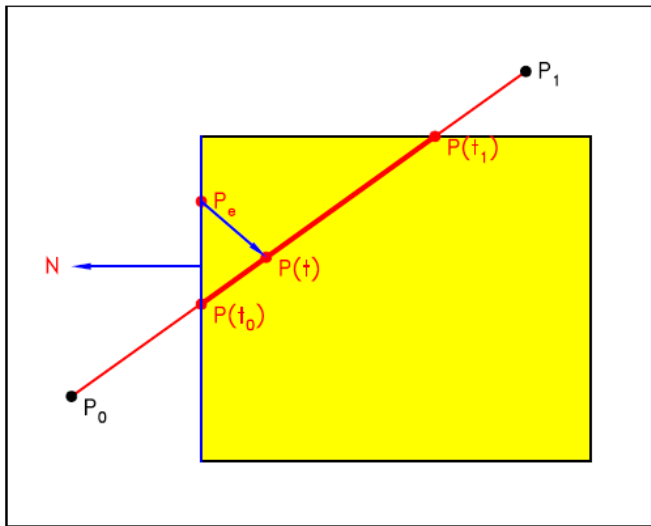
$$R : y = y_0 + (x_{max} - x_0) \cdot \frac{y_1 - y_0}{x_1 - x_0} \quad L : y = y_0 + (x_{min} - x_0) \cdot \frac{y_1 - y_0}{x_1 - x_0}$$

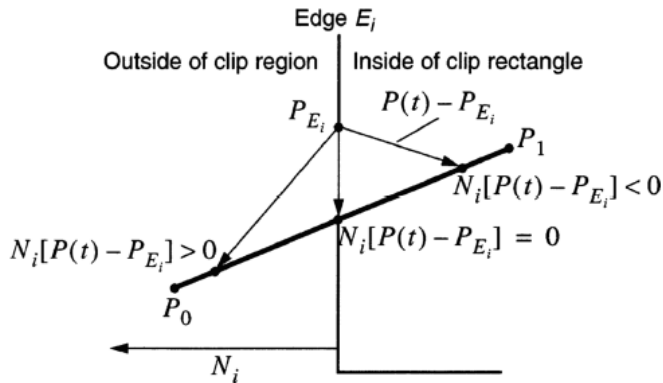
Cohen Sutherland

- Select a vertex outside
- rank window edges
- divide line at intersection with edge of highest priority
- delete line segment
- compute outcode
- iterate until trivial accept or trivial reject



Liang Barsky clipping algorithm





Let $\mathbf{P}(t) = (1 - t)\mathbf{P}_0 + t.\mathbf{P}_1$. For $\mathbf{P}(t)$ on the same edge as $\mathbf{P}_e$ and hence perpendicular to $\mathbf{N}$,

$$\begin{aligned}0 &= \mathbf{N}[\mathbf{P}(t) - \mathbf{P}_e] \\&= \mathbf{N}[(1 - t).\mathbf{P}_0 + t.\mathbf{P}_1 - \mathbf{P}_e] \\&= \mathbf{N}.\mathbf{P}_0 - t.\mathbf{N}.\mathbf{P}_0 + t.\mathbf{N}.\mathbf{P}_1 - \mathbf{N}.\mathbf{P}_e\end{aligned}$$

Bring all terms containing t to the left,

$$t.\mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_1] = \mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_e]$$

Rearrange

$$t = \frac{\mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_e]}{\mathbf{N}.[\mathbf{P}_0 - \mathbf{P}_1]}$$

$$t = \frac{\mathbf{N} \cdot [\mathbf{P}_0 - \mathbf{P}_e]}{\mathbf{N} \cdot [\mathbf{P}_0 - \mathbf{P}_1]}$$

For the left edge $\mathbf{N} = (-1, 0)$ and $\mathbf{P}_e = (x_{min}, y)$ so that

$$t = \frac{(-1, 0) \cdot (x_0 - x_{min}, y_0 - y)}{(-1, 0) \cdot (x_0 - x_1, y_0 - y_1)} = \frac{x_{min} - x_0}{x_1 - x_0}$$

For the right edge $\mathbf{N} = (1, 0)$ and $\mathbf{P}_e = (x_{max}, y)$ so that

$$t = \frac{(1, 0) \cdot (x_0 - x_{max}, y_0 - y)}{(1, 0) \cdot (x_0 - x_1, y_0 - y_1)} = \frac{x_0 - x_{max}}{x_0 - x_1}$$

$$t = \frac{\mathbf{N} \cdot [\mathbf{P}_0 - \mathbf{P}_e]}{\mathbf{N} \cdot [\mathbf{P}_0 - \mathbf{P}_1]}$$

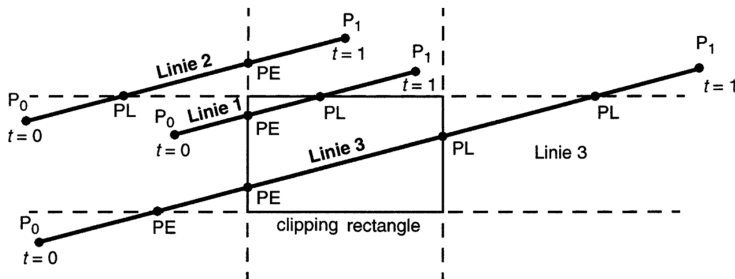
For the bottom edge $\mathbf{N} = (0, -1)$ and $\mathbf{P}_e = (x, y_{min})$ so that

$$t = \frac{(0, -1) \cdot (x_0 - x, y_0 - y_{min})}{(0, -1) \cdot (x_0 - x_1, y_0 - y_1)} = \frac{y_{min} - y_0}{y_1 - y_0}$$

For the right edge $\mathbf{N} = (0, 1)$ and $\mathbf{P}_e = (x, y_{max})$ so that

$$t = \frac{(0, 1) \cdot (x_0 - x, y_0 - y_{max})}{(0, 1) \cdot (x_0 - x_1, y_0 - y_1)} = \frac{y_0 - y_{max}}{y_0 - y_1}$$

Liang Barsky Entry and exit criteria



Entry and exit points can be found, Let $\mathbf{D} = \mathbf{P}_1 - \mathbf{P}_0$

- test for all $t \in [0, 1]$
- $\mathbf{N} \cdot \mathbf{D} < 0 \Rightarrow$ entry $\mathbf{N} \cdot \mathbf{D} > 0 \Rightarrow$ exit.
- Compute maximum (entry) $t_e = \max(t_{ei})$ and minimum (exit) $t_l = \min(t_{li})$.
- if $t_e > t_l$ no intersection.

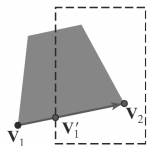
Polygon clipping

Same but for polygons.

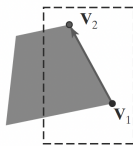
Two algorithms

- 1 Sutherland-Hodgeman algorithm
- 2 Weiler-Atherton algorithm

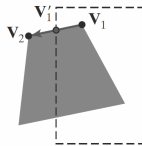
Sutherland-Hodgeman algorithm



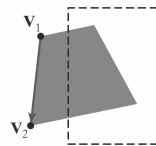
out $\rightarrow$ in
Output: V_1', V_2



in $\rightarrow$ in
Output: V_2

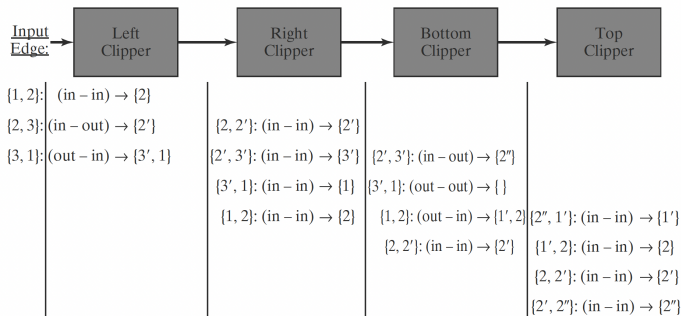
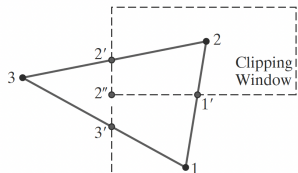


in $\rightarrow$ out
Output: V_1'

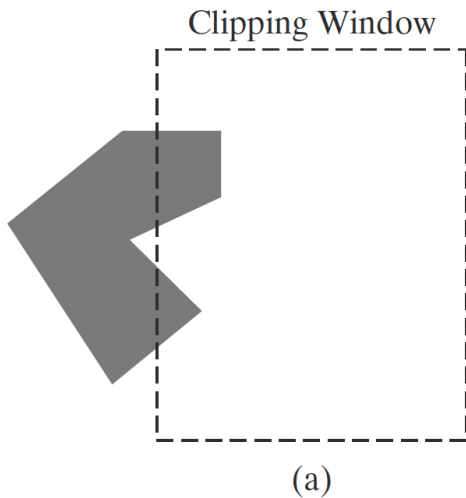


out $\rightarrow$ out
Output: none

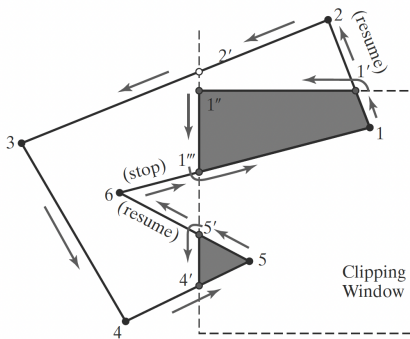
Sutherland-Hodgeman algorithm example



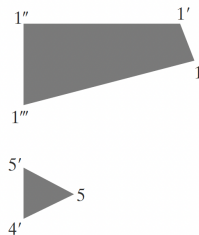
Exception



Weiler-Atherton algorithm



(a)



(b)

The End